CT-468 DEEP LEARNING, PAPER REPRODUCTION

Mimetic initialization of self-attention layers

Trockman and Kolter, ICML 2023. Reproduced on CIFAR-10 with a ViT written from scratch.

Team 404 Not Found

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The paper: copy two patterns from pretrained attention

ViTs trained from scratch on small datasets do poorly. In a ViT pretrained on ImageNet, every attention layer shows two diagonal patterns. Mimetic init builds them into the weights before training, in closed form, with no data.

Query and key, per head

$$W_Q W_K^T \approx 0.7 Z1 + 0.7 I$$

Positive diagonal: a token attends to tokens with similar input, so through the position embeddings to itself and nearby patches. A locality bias like a convolution.

Value and projection, per layer

$$W_V W_{proj} \approx 0.4 Z2 - 0.4 I$$

Negative diagonal: each attention block subtracts part of a token's own value from the residual stream, as in the pretrained layers.

Main claim, ViT 192/12/3 on CIFAR-10: **86.07% to 90.78%** (+4.71 points).

Setup: written from scratch in plain PyTorch

Data

CIFAR-10: 50,000 train and 10,000 test images, 32 x 32, 10 classes. No validation set.

Model

ViT, width 192, depth 12, 3 heads, patch 2: 257 tokens, 5.34M parameters. Fixed sinusoidal position embeddings.

Training

AdamW, lr 3e-3, weight decay 0.01, batch 512, 100 epochs, RandAugment and Cutout. One T4 on Kaggle, about 2.3 h per run.

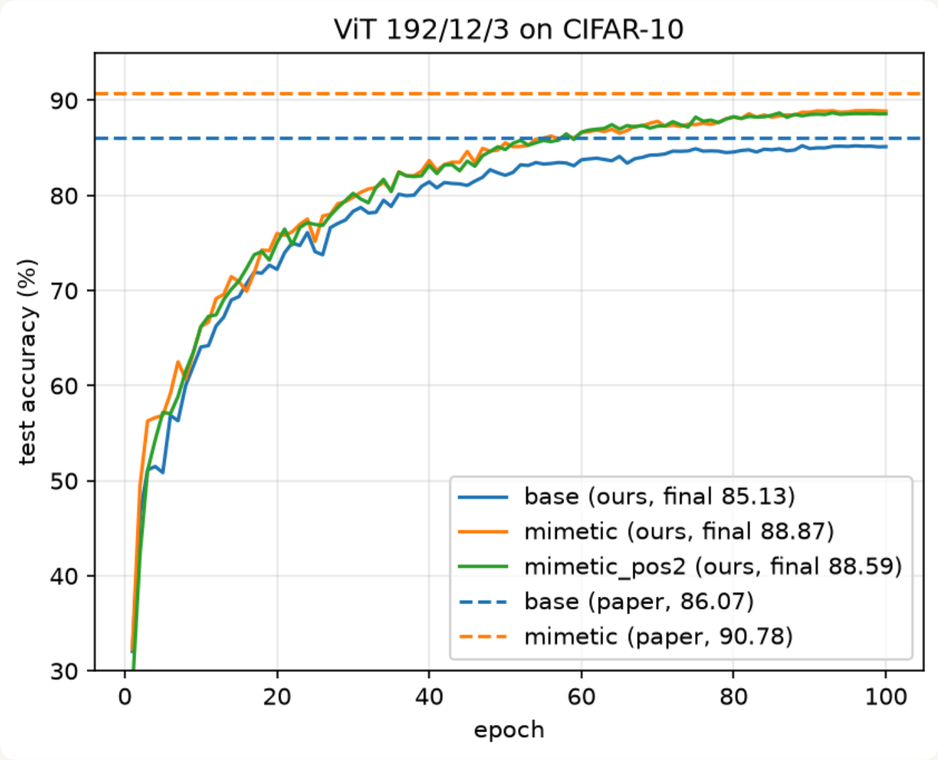
Our choices where the paper is silent

5 warmup epochs then cosine decay · RandAugment(2, 9) · Cutout 16 x 16 · random crop and flip · no dropout · float16 mixed precision · seed 0, one run per setting

Results: the claim holds, with a smaller gain

Epoch 100	Paper	Ours	Diff
Base	86.07	85.13	-0.94
Mimetic	90.78	88.87	-1.91
Gain	+4.71	+3.74	-0.97

Our gain is 79% of the paper's. Mimetic also has the smaller train-test gap: 5.93 points against 9.00.

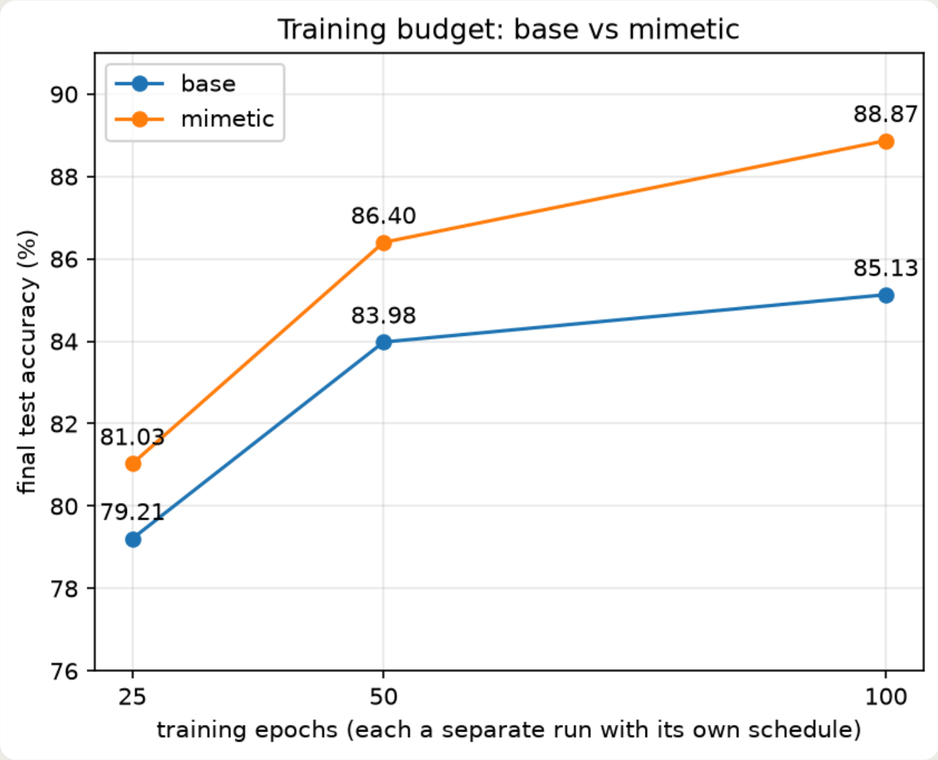


Own experiment: the gain grows with training length

Separate 25, 50 and 100 epoch runs, each with its own schedule. Only the epoch count changes.

Epochs	Base	Mimetic	Gain
25	79.21	81.03	+1.82
50	83.98	86.40	+2.42
100	85.13	88.87	+3.74

Mimetic at 50 epochs (86.40) beats base at 100 epochs (85.13).



Gap to the paper: base -0.94, mimetic -1.91

Tested: position embedding scale

The paper hints CIFAR-10 used a scale other than 1, and its Figure 4 shows about +0.5 from scale 1 to 2. One mimetic run at scale 2: 88.59 against 88.87. Not the cause.

Likely, not tested

- settings the paper does not state
- init of the other layers (paper's code base unknown)
- one run per setting

Ruled out

Under-training (flat at the end) · wrong init (self-test diagonals) · attention code (matches by hand) · best vs final epoch (at most 0.1)

Conclusions

+3.74

points from mimetic init on CIFAR-10 (paper +4.71). The claim reproduces.

50 ep

for mimetic to pass base's 100-epoch accuracy. The gain grows with training length.

1-2

points below the paper in absolute accuracy, most likely from settings the paper does not state.

A closed-form initialization with no data and no pretraining gives a large, reproducible gain for small ViTs.